Mock Project Overview: *Falcon* University Admission Pre-Assessment

+ Objective

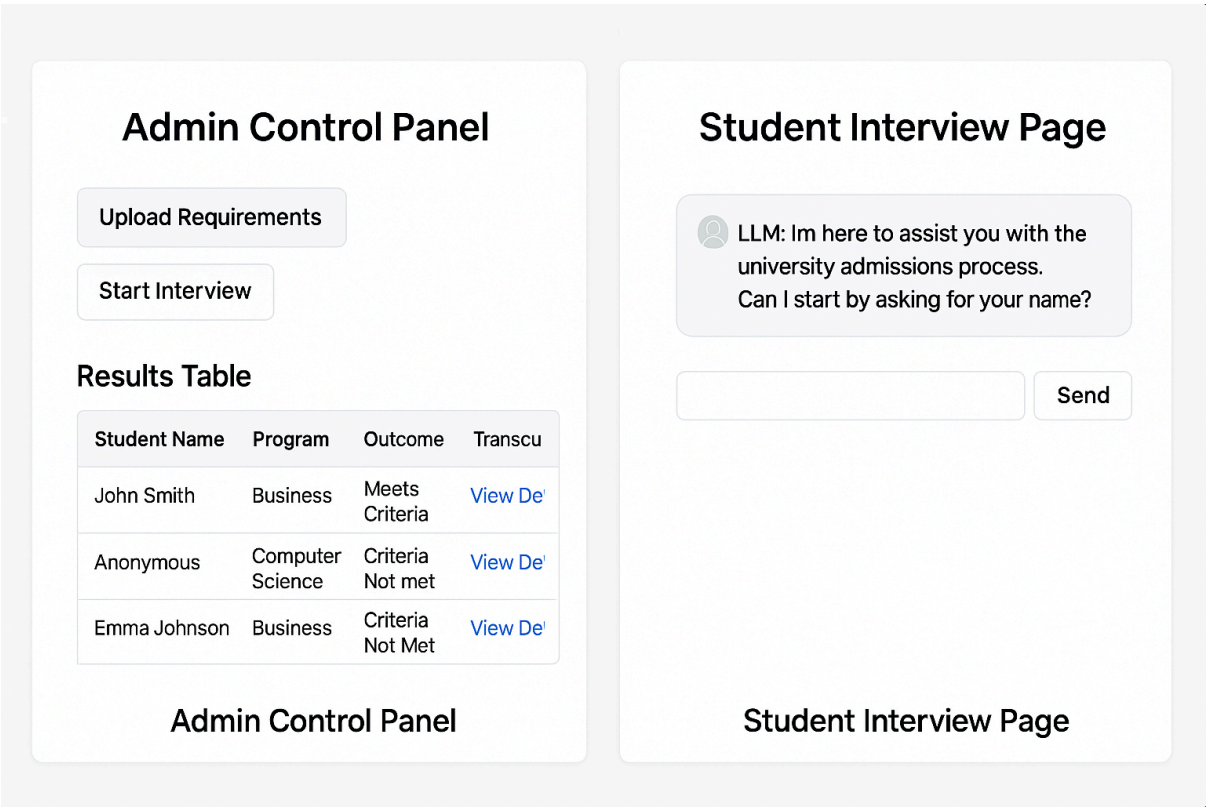
A- Build a lightweight web app that simulates an admission officer, powered by an LLM assistant, to guide applicants through a chat-based eligibility check. The bot would:

(a) chats with a student,
(b) retrieves knowledge from an uploaded PDF. (Use this)
(c) performs the eligibility check based on preset rules, and
(d) shows the outcome: Eligible to Proceed

B- After finishing step A, [lease think about the product [ ] and answer the product thinking questions..

⏱️ **Deadline:** Wednesday, **04/22/2025**

+ Workflow



Step 1: Build Admin page

1. **Login Page with Authentication**
2. **Dashboard Page** → Includes
    1. *"Upload Requirements" Button*
    2. Result Table (check next section)
    3. *"Start interview" Button*

3. **Admin uploads <u>a PDF file</u> that includes the chatbot's questions and requirements** → Use knowledge base as exmple→ Once uploaded, the backend parses text and prepares it as the knowledge base for chat.

4. Create the applicants database

   Column Description:

   - Student Name Name entered during interview (or "Anonymous" if not provided).
   - Program Program applied for (Business or Computer Science)
   - Outcome Final eligibility result: Meets Criteria / Criteria Not Met

- Rule Summary Short text explanation
- Transcript Link Button or link to open the View Details drawer with chat transcript.
- Transcription Conversation between LLM and user

Step 2: Build User page

1. **"Start Interview"** Button→ Clicking the button opens the chat panel.
2. **LLM Q&A and Eligibility Check** →
    - LLM greets the student and begins asking eligibility questions.
    - Student replies in text.
    - Once data is collected, the LLM evaluates the rules, tells the student the result, and closes the chat.

Step 3: Update the admin page with the user's input

1. **Table Updates** → The results table updates with the gathered information

Step 4: Deployment

Deployment options:

i. *Local*
ii. *Bonus:* *Deploy webapp on AWS or other cloud providers free accounts (Vercel,Railway,etc)

+ Suggested Stack

- **Frontend**: Next.js or Vite + Tailwind + Vercel

- **Backend**: Node.js , FastAPI (Python)
- **LLM**: OpenAI and Lang Chain (RAG + tool/function-calling)
- **Database**: PostgreSQL, or AWS stacks (Bonus)
- **Dev/Deploy**: Docker Compose for local; AWS for optional cloud deployment(Bonus)

+ Evaluation Criteria

- Functionality (chat + eligibility check working end-to-end).
- Code quality (clean structure, modular backend).
- LLM/RAG use (retrieval applied meaningfully).
- UI/UX (simple, clean dashboard; transcript and reasons visible).

+ Submission Requirements

- Share a GitHub repo with https://github.com/rmyzm
- Short Loom demo (2–3 minutes: demo and explanation
    - **Showcase the demo**
    - **Mention the tech stack you used**
    - **Explain where and how the LLM or Copilots were used to build the webpage**
- README with setup instructions.
- README includes responses to product thinking questions.